# Italian Road Safety: Where to Invest First

Which Italian municipalities are the best ones for our company to invest in, and why?

# Objective & Implementation Plan

## The assignment

An updated dashboard of the best Italian municipalities to invest in: highest crash rates, total crashes, population and any statistic that matters to the business.

Optional: a forecast (not included in this project). The available data was unreliable, outdated and too dirty to work with.

## Analysis workflow

- Fetch accident data from the ISTAT API.
- Clean data and join SITUAS information.
- Verify coverage for 2021–2024.
- Explore trends and calculate municipal indicators.
- Apply clustering, hypothesis testing and regression.
- Build a priority score and Power BI dashboard.

# Data preparation

| What I Did | Why |
|---|---|
| Downloaded ISTAT data via API and examined variables and codes. | To understand how crashes, injuries and deaths are recorded. |
| Reshaped the data and checked duplicates, missing values, national totals and consistency. | To verify the dataset before analysis. |
| Standardised municipality codes, cleaned SITUAS numeric formats and joined the sources. | To enrich accident records with names, population and area. |
| Investigated extreme values, unmatched codes and changes in annual coverage. | To distinguish real observations from data-quality issues. |
| Selected complete 2021–2024 records and calculated annual averages and rates. | To compare municipalities over a consistent period. |
| Restricted rankings to municipalities with 10,000+ residents. | To reduce instability in rates for very small towns. |

Checks: no duplicate municipality-year records; injured + deaths ≥ crashes; four-year coverage verified. Final comparison dataset: 6,331 municipalities, including 1,206 with 10,000+ residents. Population and area refer to 2020.

# Analysis: four methods for four questions

## Clustering

**Which municipalities look alike?**

K-means on crash rate, crash density and population, after log transformations and standardisation. Two groups selected by comparing silhouette scores for 2–6 clusters.

## Hypothesis test

**Do crash-rate distributions differ across macro-areas**

Mann–Whitney test compares municipal crash-rate distributions. Median: 2.61 in North–Centre versus 1.80 in South–Islands; $p < 0.001$.

## Regression

**Is it only size and density?**

Exploratory OLS relates crash rates to population, population density and macro-area. $R^2 = 0.319$; HC3 robust standard errors. Associations, not causes.

## Priority score

**Which municipalities should we investigate first?**

Equal-weight average of crash-rate and crash-density percentiles. Alternative weights retain 13 of the top 15 municipalities; the same five remain at the top.

Before these: exploratory charts on trends, distributions and outliers. Results are on the dashboard.

# Tools and final recommendation

| | |
|---|---|
| **Data access** | ISTAT API<br>SITUAS |
| **Preparation** | Python: requests, pandas, numpy |
| **Statistics and ML** | scikit-learn, scipy, statsmodels |
| **Charts** | matplotlib (EDA), Power BI (final story) |
| **Versioning** | Git and GitHub: notebook, data, dashboard, deck |

## What we recommend

Prioritise further assessment in Bergamo, Florence, Genoa, Milan and Riccione. Compare crash volumes and fatalities, then collect traffic data and estimate intervention costs before allocating funds.

The score identifies investigation priorities, not expected investment returns. Population and area refer to 2020; traffic exposure is not measured.

The notebook runs from data fetching to final tables; the dashboard reads the exported files.